

Standing Seam Roof Detail Notes

Technical notes for seams, concealed fastening, roof movement, flashing connections, and drainage coordination.

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What Makes Standing Seam Different

Standing seam roofing waterproofs through a raised, interlocked seam and hides its fasteners, so the weather line is never punctured. This changes how the roof handles movement, span, and slope compared with screw-fixed systems.

Aspect	Standing Seam	Sandwich Panel	Corrugated
Waterproofing	Raised interlocked seam	Lapped joint + sealant	Lapped profile + sealant
Fastening	Concealed clips	Exposed screws	Exposed screws
Thermal movement	Free (floating clips)	Restrained at fixings	Restrained at fixings
Span capability	Long, continuous	Medium	Short-medium
Typical slope	From 1°	From 3°	From 5°
Best use case	Long-run, low-slope roofs	Insulated envelopes	Low-cost shelters

Critical Detail 1: Seam Types

Snap-lock

Panels click together by hand; no seaming tool needed. Fast to install and ideal for moderate slopes and lower wind zones.

Mechanically seamed (single / double lock)

An electric seamer folds the seam 90° or 180°. Highest wind-uplift resistance; preferred for low slopes and exposed sites — recommended for windy plateau and coastal zones in Iran.

Zip-lock

A T-shaped seam snaps over a concealed clip. Good balance of speed and performance for medium spans.

Seam type	Tool required	Uplift resistance	Recommended for
Snap-lock	None	Medium	Moderate slope, sheltered sites
Mechanically seamed	Electric seamer	High	Low slope, exposed/windy sites
Zip-lock	None	Medium-high	Medium spans, general use

Critical Detail 2: Thermal Movement Management

Metal roofs expand and contract with temperature. The movement must be allowed to occur freely, or seams distort and fasteners elongate their holes.

Calculation method

$\Delta L = L \times \alpha \times \Delta T$ — where L = panel length, α = thermal expansion coefficient (steel $\approx 12 \times 10^{-6}$ /°C), ΔT = temperature swing.

Worked example

30 m panel $\times 12 \times 10^{-6} \times 60^\circ\text{C} = 21.6$ mm of movement along the panel.

Design rules

- Locate one fixed point per panel; the rest of the clips float.
- Floating clip spacing at maximum 600 mm centers.
- Add anti-drumming tape or closer clip spacing on long, exposed runs.
- Allow movement at eave, ridge, and penetration flashings.

Critical Detail 3: Flashing at Eave, Ridge & Wall

Eave

Anti-capillary groove formed at the panel end; minimum 50 mm overhang into the gutter so water discharges cleanly and cannot track back.

Ridge

Double-locked cap with foam closure and sealant at both sides; the cap must allow thermal movement of the panels beneath it.

Wall (abutment)

Z-flashing with a minimum 150 mm lap and a sealed top edge; the upstand turns up behind the wall cladding to keep the lap line dry.

- Match flashing gauge and coating to the roof sheet.
- Never face-fix through the weather line on a standing seam roof.
- Use butyl or neutral-cure sealant compatible with the coating.

Installation Sequence & Common Errors

Installation sequence

1. Confirm purlin line and level; correct any high/low purlins.
2. Set and fix the eave flashing and starter.
3. Establish the fixed point and square the first panel.
4. Set the first panel to the gable reference line.
5. Install concealed clips at design spacing.
6. Engage and seam each successive panel.
7. Seam mechanically where specified (single/double lock).
8. Form valley and penetration flashings as panels progress.
9. Fit ridge closure and double-locked ridge cap.
10. Walk the roof, check seams, and clean off all swarf.

Common errors

- Pinning every clip — kills thermal movement and buckles seams.
- No anti-capillary groove at eave — water tracks back under the sheet.
- Ridge cap face-fixed — punctures the weather line.
- Clip spacing too wide on exposed sites — seam unzips under uplift.
- Swarf left on the roof — rusts and stains the coating.

For seam selection and uplift design on your project, contact SIPANEL engineering:
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